

```

1 #%%
2 import pandas as pd
3 import numpy as np
4
5 df = pd.read_csv('PS/train.csv')
6 #%%
7
8 suspect_cols = [
9     'Kharif Seasons Agro Ecological Sub Zone in 2022
10     ',
11     'Rabi Seasons Agro Ecological Sub Zone in 2022',
12     'Rabi Seasons Agro Ecological Sub Zone in 2021',
13     'Kharif Seasons Agro Ecological Sub Zone in 2021'
14     ',
15     'Kharif Seasons Agro Ecological Sub Zone in 2020'
16     ',
17     'Rabi Seasons Agro Ecological Sub Zone in 2020'
18 ]
19 string_correlation_matrix = df[suspect_cols].apply(
20     lambda x: x.astype('category').cat.codes).corr()
21 #%%
22 suspect_cols = [
23     'Kharif Seasons Type of soil in 2022',
24     'Rabi Seasons Type of soil in 2022',
25     'Rabi Seasons Type of soil in 2021',
26     'Kharif Seasons Type of soil in 2021',
27     'Kharif Seasons Type of soil in 2020',
28     'Rabi Seasons Type of soil in 2020'
29 ]
30 string_correlation_matrix = df[suspect_cols].apply(
31     lambda x: x.astype('category').cat.codes).corr()
32 #%%
33 suspect_cols = [
34     'Kharif Seasons Type of water bodies in hectares
35     2022',

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```
36     'Rabi Seasons Type of water bodies in hectares
    2022',
37     'Rabi Seasons Type of water bodies in hectares
    2021',
38     'Kharif Seasons Type of water bodies in hectares
    2021',
39     'Kharif Seasons Type of water bodies in hectares
    2020',
40     'Rabi Seasons Type of water bodies in hectares
    2020'
41 ]
42
43 string_correlation_matrix = df[suspect_cols].apply(
    lambda x: x.astype('category').cat.codes).corr()
44
45 print(string_correlation_matrix)
```